Unit 11: Problem on Ages and Numbers

CONTENTS

Objectives

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- We will understand how to calculate age some years ago
- We will understand how to calculate present age
- We will understand how to calculate age some years hence

Introduction

Problems based on ages are generally asked in most of the competitive examinations. To solve these problems, the knowledge of linear equations is essential. In such problems, there may be three situations:

(i) Age some years ago

(ii) Present age

(iii) Age some years hence

Two of these situations are given and it is required to find the third. The relation between the age of two persons may also be given. Simple linear equations are framed and their solutions are obtained. Sometimes, short cut methods given below are also helpful in solving such problems.

11.1 Shortcut Method

1. If the age of A, t years ago, was n times the age of B and at present A's age is n_2 times that of B, then

A's present age $= \left(\frac{n_1-1}{n_1-n_2}\right) n_2 t$ Years

n_2t years and, B's present age = $\left(\frac{n_1-1}{n_1-n_2}\right) t$ years

Explanation:-

Let the present age of B be x years. Then, the present age of A = n_2x years

Given, t years ago,

$n_1(x - t) = n_2x - t$ or, $(n_1 - n_2)x = (n1 - 1)t$

or X = $\left(\frac{n_1-1}{n_1-n_2}\right)t$

Therefore, B's present age = $\left(\frac{n_1-1}{n_1-n_2}\right)$t years

And A's present age = $\left(\frac{n_1-1}{n_1-n_2}\right)$ n_2t years.

Example 1: The age of father is 4 times the age of his son. If 5 years ago father's age was 7 times the age of his son at that time, then what is father's present age?

Solution: The father's present age

= $\left(\frac{n_1-1}{n_1-n_2}\right) n_2t$ [Here, n_1 = 7, n_2 = 4 and t = 5]

= $\left(\frac{7-1}{7-4}\right)$ 4 x 5 = $\frac{6\,x\,4\,x\,5}{3}$

= 40 years

2) The present age of A is n_1 times the present age of B. If t years hence, the age of A would be n_2 times that of B, then

And A's present age = $\left(\frac{n_1-1}{n_1-n_2}\right)$ n_2t years.

And And B's present age = $\left(\frac{n_1-1}{n_1-n_2}\right)$ t years.

2) The present age of A is n_1 times the present age of B. If t years hence, the age of A would be n_2 times that of B, then

And A's present age = $\left(\frac{n_1-1}{n_1-n_2}\right)$ n_2t years.

And And B's present age = $\left(\frac{n_1-1}{n_1-n_2}\right)$ t years.

Explanation

Let the present age of B be x years.

Then, the present age of A = n_1x

Given, t years hence,

(n_1x + t) = n_2 (x + t)

or, (n_1 - n_2)x = (n_2 - 1)t

Or $\left(\frac{n_1-1}{n_1-n_2}\right)$ t years.

Therefore B's present age = $\left(\frac{n_2-1}{n_1-n_2}\right)$ n_1t years.

And A's present age = $\left(\frac{n_2-1}{n_1-n_2}\right)$ n_1t years.

Example 2: The age of Mr Gupta is four times the age of his son. After ten years, the age of Mr Gupta will be only twice the age of his son. Find the present age of Mr Gupta's son.

Solution: The present age of Mr Gupta's son

= $\left(\frac{n_2-1}{n_1-n_2}\right)$ t years.

= $\left(\frac{2-1}{4-2}\right)$ 10

[Here, n_1 = 4, n_2 = 2 and t = 10]

= 5 years.

3) The age of A, t_1 years ago, was n_1 times the age of B. If t 2 years hence A's age would be n2 times that of B, then

A' s present age = $\frac{n_1(t_1+t_2)(n_2-1)}{n_1-n_2} + t_1$ years

B' s present age = $\frac{t_2(n_2+1)+t_1(n_1-1)}{n_1-n_2}$ years

Explanation:-Let A's present age = x years and B's present age = y years.

Given: x - t_1 = n_1 (y - t_1) and x + t_2 = n_2 (y + t_2) i.e., x - n_1 y = (1 - n_1) t_1 (1)

and, x - n_2 y = (-1 + n_2) t_2 (2)

Solving (1) and (2), we get

X= $\frac{n_1(t_1+t_2)(n_2-1)}{n_1-n_2} + t_1$

And ,y = $\frac{t_2(n_2+1)+t_1(n_1-1)}{n_1-n_2}$

Example 3: 10 years ago Anu's mother was 4 times older than her daughter. After 10 years, the mother will be twice older than her daughter. Find the present age of Anu is:

Solution: Present age of Anu

= $\frac{t_1(n_2-1)+t_1(n_1-1)}{n_1-n_2}$ years

[Here, n_1 = 4, n_2 = 2, t_1 = 10 and t_2 = 10]

= $\frac{10(2-1)+10(4-1)}{4-2}$

= $\frac{10+30}{2}$

= 20 years

4) The sum of present ages of A and B is S years. If, t years ago, the age of A was n times the age of B, then

Present age of A =$\frac{Sn-t(n-1)}{n-1}$ years

And present age of B = $\frac{S-t(n-1)}{n-1}$ years

Explanation

Let the present ages of A and B be x and y years

respectively.

Given: x + y = S (1)

and, x - t = n (y - t)

or, x - ny = (1 - n)t (2)

Solving (1) and (2), we get

X = $\frac{Sn-t(n-1)}{n-1}$

Y = $\frac{S-t(n-1)}{n-1}$ years

Example 4: The sum of the ages of A and B is 42 years. 3 years back, the age of A was 5 times the age of B. Find the difference between the present ages of A and B.

Solution: Here, S = 42, n = 5 and t = 3

∴ Present age of A

= $\frac{Sn-t(n-1)}{n-1}$

= $\frac{42 \, x \, 5-3(5-1)}{5+1}$

= $\frac{198}{6}$

= 33 years

And, present age of B

$= \frac{42+t(n+1)}{n+1}$

$= \frac{42+3(5-1)}{5+1}$

$= \frac{54}{6}$

= 9 years

∴ Difference between the present ages of A and B = 33 - 9 = 24 years.

6) The sum of present ages of A and B is S years. If, t years hence, the age of A would be n times the age of B, then

Present age of A $= \frac{Sn-t(n-1)}{n-1}$ years

And present age of B = $\frac{S-t(n-1)}{n-1}$ years

Explanation:-

Let the present ages of A and B be x and y years,

respectively

Given: x + y = S (1)

and, x + t = n (y + t)

or, x - ny = t (n - 1) (2)

Solving (1) and (2), we get

X = $\frac{Sn-t(n-1)}{n-1}$ years

Y = $\frac{S-t(n-1)}{n-1}$

Example 5: The sum of the ages of a son and father is 56 years. After four years, the age of the father will be three times that of his son. Find their respective ages.

Solution: The age of the father

$= \frac{S_n+t(n-1)}{n+1}$

$= \frac{56x\ 3+4(3-1)}{3+1}$

[Here, S = 56, t = 4 and n = 3]

$= \frac{176}{4}$

= 44 years

6) If the ratio of the present ages of A and B is a:b and t years hence, it will be c:d, then

A' s present age = $\frac{at(c-d)}{ad-bc}$

And, B's present age = $\frac{bt(c-d)}{ad-bc}$

Example 6: The ratio of the age of father and son at present is 6:1. After 5 years, the ratio will become 7:2. Find the present age of the son.

Solution: The present age of the son = $\frac{bt(c-d)}{ad-bc}$

[Here, a = 6, b = 1, c = 7, d = 2 and t = 5]

$= \frac{1\ x\ 5(7-2)}{6x2-1x7}$

= 5 years

Example 7: 6 years ago Mahesh was twice as old as Suresh. If the ratio of their present ages is 9:5 then, what is the difference between their present ages?

Solution: Present age of Mahesh

$=\frac{-at(c-d)}{ad-bc}$

$= \frac{-9x6(2-1)}{1x9-5x2}$

[Here, a = 9, b = 5, c = 2, d = 1 and t = 6]

= 54 years

Present age of Suresh

$= \frac{bt(c-d)}{ad-bc}$

$= \frac{-5x6(2-1)}{1x\,9-5x2}$

= 30 years

Difference of their ages = 54 – 30 = 24 years

Example1. 10, years ago, Mohan was thrice as old as Ram was but 10 years hence, he will be only twice as old as Ram. Find Mohan's present age.

(a) 60 years

(b) 80 years

(c) 70 years

(d) 76 years

Ans: (c) Let, Mohan's present age be x years and Ram's present age be y years.

Then, according to the fi rst condition,

x − 10 = 3(y − 10)

or, x − 3y = −20 …(1)

Now, Mohan's age after 10 years

= (x + 10) years Ram's age after 10 years

= (y + 10)

∴ (x + 10) = 2 (y + 10)

or, x − 2y = 10 … (2)

Solving (1) and (2), we get

x = 70 and, y = 30

∴ Mohan's age = 70 years and Ram's age = 30 years.

Example 2. The ages of Ram and Shyam differ by 16 years. 6 years ago, Mohan's age was thrice as that of Ram's, fi nd their present ages.

(a) 14 years, 30 years

(b) 12 years, 28 years

(c) 16 years, 34 years

(d) 18 years, 38 years

Ans: (a) Let, Ram's age = x years So, Mohan's age = (x + 16) years Also, 3(x − 6) = x + 16 − 6 or, x = 14

∴ Ram's age = 14 years and, Mohan's age = 14 + 16 = 30 years.

Example 3. 15 years hence, Rohit will be just four times as old as he was 15 years ago. How old is Rohit at present?

(a) 20

(b) 25

(c) 30

(d) 35

Ans: (b) Let, the present age of Rohit be x years Then, given: x + 15 = 4(x − 15) ⇒ x = 25.

Example 4. A man's age is 125% of what it was 10 years ago, but 83 $\frac{1}{3}$% of what it will be after ten 10 years. What is his present age?

(a) 45 years

(b) 50 years

(c) 55 years

(d) 60 years

Ans: (b) Let, the present age be x years.

Then, 125% of (x − 10) = x

and, 83 $\frac{1}{3}$ % of (x + 10) = x

∴ 125 % of (x − 10) = 83 $\frac{1}{3}$ % of (x + 10)

or, $\frac{5}{4}$ (x − 10) = $\frac{5}{6}$ (x + 10)

or, $\frac{5}{4}x - \frac{5}{6}x = \frac{50}{6} + \frac{50}{4}$

or, $\frac{5x}{12} = \frac{250}{12} or, x = 50 years.$

Example 5. If twice the son's age be added to the father's age, the sum is 70 years and if twice the father's age is added to the son's age, the sum is 95 years. Then father's age is:

(a) 40 years

(b) 35 years

(c) 42 years

(d) 45 years

Ans: (a) Let, son's age (in years) = x and father's age (in years) = y Given: 2x + y = 70 and, x + 2y = 95 Solving for y, we get y = 40.

Example 6. 3 years ago, the average age of a family of 5 members was 17 years. A baby having been born, the average age of the family is the same today? What is the age of the child?

(a) 3 years

(b) 5 years

(c) 2 years

(d) 1 year

Ans: (c) Present age of 5 members = 5 × 17 + 3 × 5 = 100 years Also, present ages of 5 members + Age of the baby = 6 × 17 = 102 years ∴ Age of the baby = 102 − 100 = 2 years.

Example 7. The ratio of A's and B's ages is 4:5. If the difference between the present age of A and B 5 years hence is 3, then what is the sum of present ages of A and B?

(a) 68 years

(b) 72 years

(c) 76 years

(d) 64 years

Ans: (b) Given: $\frac{A}{B}$= 4 5or, B = $\frac{5}{4}$ A

and, B − (A + 5) = 3 or, B = A + 8

∴ $\frac{5}{4}$ A = A + 8

or, $A\left(\frac{5}{4} - 1\right) = 8$

∴ A = 32 years and, B = $\frac{5}{4}$ × 32 = 40 years

∴ A + B = 40 + 32 = 72 years.

Example 8. The ages of A and B are in the ratio 6:5 and sum of their ages is 44 years. The ratio of their ages after 8 years will be:

(a) 4:5

(b) 3:4

(c) 3:7

(d) 8:7

Ans: (d) Let, present ages (in years) of A and B respectively, are 6x and 5x. Given: 6x + 5x = 44 ⇒ x = 4

Ratio of ages after 8 years will be 6x + 8:5x + 8 or, 32:28 or, 8:7.

Example 9. One year ago the ratio between Samir and Ashok's age was 4:3. One year hence the ratio of their ages will be 5:4. What is the sum of their present ages in years?

(a) 12 years

(b) 15 years

(c) 16 years

(d) Cannot be determined

Ans: (c) Let, one year ago

Samir's age be 4x years

and, Ashok's age be 3x years

Present age of Samir = (4x + 1) years

Present age of Ashok = (3x + 1) years

One year hence

Samir's age = (4x + 2) years

Ashok's age = (3x + 2) years

According to question, $\frac{4x+2}{3x+2} = \frac{5}{4}$ ⇒16x + 8 = 15x + 10

or, x = 2.

∴ Sum of their present ages = 4x + 1 + 3x + 1

= 7x + 2

= 7 × 2 + 2 = 16 years.

Example 10. Ratio of Ashok's age to Pradeep's age is 4:3. Ashok will be 26 years old after 6 years. How old is Pradeep now?

(a) 18 years

(b) 21 years

(c) 15 years

(d) 24 years

Ans: (c) Let, the present ages of Ashok and Pradeep be 4x and 3x

So that $4x + 6 = 26 \Rightarrow x = 5$

∴ Present age of Pradeep is $3x = 3 \times 5$, i.e., 15 years

Example 11. Jayesh is as much younger to Anil as he is older to Prashant. If the sum of the ages of Anil and Prashant is 48 years, what is the age of Jayesh?

(a) 20 years

(b) 24 years

(c) 30 years

(d) Cannot be determined

Ans: (b) A – J = J – P

2J = A + P

2J = 48

= J 24 years

Example 12. 5 years ago, Mr Sohanlal was thrice as old as his son and 10 years hence he will be twice as old as his son. Mr Sohanlal's present age (in years) is:

(a) 35

(b) 45

(c) 50

(d) 55

Ans: (c) Let, Mr Sohanlal's age (in years) = x and his son's age = y Then, $x - 5 = 3 (y - 5)$ i.e., $x - 3y + 10 = 0$ and, $x + 10 = 2 (y + 10)$ i.e., $x - 2y - 10 = 0$ Solving the two equations, we get $x = 50, y = 20$.

Example 13. Three times the present age of a father is equal to eight times the present age of his son. 8 years hence the father will be twice as old as his son at that time. What are their present ages? (a) 35, 15

(b) 32, 12

(c) 40, 15

(d) 27, 8

Ans: (b) 3F = 8S ⇒ F 8/3S

F + 8 = 2 (S + 8)

F + 8 = 2S + 16

$\frac{8}{3}$S +8= 2S +16

8S + 24 = 6S + 48

2S = 24

S= 12

F =8/3 ×S

=8/3 × 12⇒ 32

Example 14. The sum of the ages of a father and son is 45 years. 5 years ago, the product of their ages was four times the father's age at that time. The present age of the father is:

(a) 39 years

(b) 36 years

(c) 25 years

(d) None of these

Ans: (b) Let, father's present age = x years Then, son's present age = (45 - x) years Given: (x - 5)(45 - x - 5) = 4(x - 5) or, x^2 - 41x + 180 = 0 or, (x - 36)(x - 5) = 0 ∴ x = 36 years.

Example 15. One year ago, a father was four times as old as his son. In 6 years time his age exceeds twice his son's age by 9 years. Ratio of their ages is:

(a) 13:4

(b) 12:5

(c) 11:3

(d) 9:2

Ans: (c) Let, the present ages of father and son be x and y years, respectively Then, (x - 1) = 4(y - 1) or, 4y - x = 3 …(1)

and, (x + 6) - 2(y + 6) = 9 or, -2y + x = 15 …(2)

Solving (1) and (2), we get, x = 33, y = 9

∴ Ratio of their ages = 33:9 = 11:3.

Example 16. The ages of A, B and C together is 185 years. B is twice as old as A and C is 17 years older than A. Then, the respective ages of A, B and C are:

(a) 40, 86 and 59 years

(b) 42, 84 and 59 years

(c) 40, 80 and 65 years

(d) None of these

Ans: (b) Let, A's age be x years B's age be 2x years C's age = (x + 17) years According to the question, x + 2x + (x + 17) = 185 ∴ 4x = 185 - 17 = 168

∴ x = 42

∴ A's age = 42 years B's age = 84 years C's age = 42 + 17 = 59 years.

Example 17. A father's age is three times the sum of the ages of his two children, but 20 years hence his age will

be equal to the sum of their ages. Then, the father's age is:

(a) 30 years

(b) 40 years

(c) 35 years

(d) 45 years

Ans: (a) Let, the present age of father be x years and the present age of son be y years. ∴ x = 3y … (1)

Also, (x + 20) = (y + 20 + 20) … (2)

Solving (1) and (2), we get x = 30 years.

EXERCISE 2

Example 1. Eighteen years ago, the ratio of A's age to B's age was 8:13. Their present ratios are 5:7. What is the present age of A?

(a) 70 years

(b) 50 years

(c) 40 years

(d) 60 years

Ans: (b) Let A → age of A (Present)

B → age of B (Present)

$$\frac{A-18}{B-18} = {}^{8}/_{13}$$

$$\frac{\frac{5}{7}B-18}{B-18} = {}^{8}/_{13}$$

$$\frac{5B-126}{7B-126} = {}^{8}/_{13}$$

65B – 1638 = 56B – 1008

9B = 630

B=70

A=$\frac{5}{7} \times B$

=$\frac{5}{7} \times 70 = 50 year.$

Example 2. The average age of 30 students of a class is 14 years 4 months. After admission of 5 new students in the class the average becomes 13 years 9 months. The youngest one of the fi ve new students is 9 years 11 months old. The average age of the remaining 4 new students is

(a) 12 years 4 months

(b) 11 years 2 months

(c) 10 years 4 months

(d) 13 years 6 months

Ans: (c) Let a be the age of youngest student.

According to the question,

$$30\left(14\frac{1}{3}\right) + 9\frac{11}{2} + 4a = 35 \times 13 + \frac{35 \times 3}{4}$$

∴a=10 $^1/_3$

∴= 10 years 4 months.

Example 3. The sum of the ages of two brothers, having a difference of 8 years between them, will double after 10 years. What is the ratio of the age of the younger brother to that of the elder brother?

(a) 8: 9

(b) 10: 13

(c) 7: 11

(d) 3: 7

Ans: (d) 1^{st} brother → A (present age)

2nd brother → B (present age)

Given ⇒A-B=8year (1)

Sum of their age will double after 10 years

∴ A+10+B+10=2(A+B)

A+B=20 (2) From (1) and (2) ⇒ $A = 14\ B = 6 \therefore \frac{B}{A} = \frac{6}{14} = {}^3/_7$

Example 4. After replacing an old member by a new member, it was found that the average age of fi ve members of a club is same as it was 3 years ago. The difference between the ages of the replaced and the new members is:

(a) 2 years

(b) 4 years

(c) 8 years

(d) 15 years

Ans: (d) Increase in ages of five members in 3 years = (3 × 5) years = 15 years

Since the average age remains same, therefore, required difference = 15 years

Example 5. The ratio of the present ages of A and B is 7:9. 6 years ago the ratio of $\frac{1}{3}$ of A's age at that time and $\frac{1}{3}$of B's ageat that time was 1:2. What will be the ratio of A's to B's age 6 years from now?

(a) 4:5

(b) 14:15

(c) 6:7

(d) 18:25

(e) 22:25

Ans: (c) Let, the present age of A be 7x years and that of B be 9x years.

Now, 6 years ago,$\frac{3(7x-6)}{3(9x-6)} = \frac{1}{2}$

or, 42x - 36= 27x - 18 or, 15x = 18

$\therefore$x = $\frac{6}{5}$ years

Ratio after 6 years

$$\frac{\frac{7\times 6}{5}+6}{9\times\frac{6}{5}+6} = \frac{42+30}{54+30} = \frac{72}{84} = 6:7$$

$\therefore$ Required ratio = 6:7

Example 6. The present age of Romila is 1 4 that of her father. After 6 years her father's age will be twice the age of Kapil. If Kapil celebrated fi fth birthday 8 years ago, what is Romila's present age?

(a) 7 years

(b) 7.5 years

(c) 8 years

(d) 8.5 years

(e) None of these

Ans: (c) Kapil's present age = (8 + 5) = 13 years

Kapil's age after 6 years =13 + 6 = 19 years

Now, Romila's father's age = 2 × Kapil's age = 2 × 19 = 36 years

Father's present age = 38 − 6 = 32 years

Romila's present age =$\frac{1}{4}$ × father's present age =× = $\frac{1}{4}$ ×32 8years

Example 7. The average age of women and child workers in factory was 15 years. The average age of all the 16 children was 8 years and the average age of women workers was 22 years. If 10 women workers were married, then the number of unmarried women workers is:

(a) 16

(b) 12

(c) 8

(d) 6

Ans: (d) Let, unmarried women workers are x, then as per question,$\frac{16\times8+22\times(10+x)}{16+10+x} = 15$

⇒ 128 + 220 + 22x = 390 + 15x ⇒ 7x = 42

∴ x = 6

Example 8. The age of a father is three times of that of his son. After 5 years, the double of father's age will be fi ve times the age of son. The present age of father and son is:

(a) 30 years, 10 years

(b) 36 years, 12 years

(c) 42 years, 14 years

(d) 45 years, 15 years

Ans: (d) Let, present age of son is x years and then present age of father is 3x years then, $5 (x + 5) = 2 (3x + 5) \Rightarrow 5x + 25 = 6x + 10$

$\therefore x = 15$ years Present age of father = 45 years.

Example 9. The sum of the ages of 4 members of a family, 5 years ago, was 94 years. Today, when the daughter has been married off and replaced by a daughter-in-law, the sum of their ages is 92. Assuming that there has been no other change in the family structure and all the people are alive, what is the difference between the age of the daughter and that of the daughter-in-law?

(a) 22 years

(b) 11 years

(c) 25 years

(d) 19 years

(e) 15 years

Ans: (a) There are four members in a family. Five years ago the sum of ages of the family members = 94 years Now, sum of present ages of family members = $94 + 5 \times 4 = 114$ years $\therefore$ Daughter is replaced by daughter-in-law. Thus, sum of family member's ages becomes 92 years. $\therefore$ Difference = $114 - 92 = 22$ years

Example 10. The ratio between the present age of Manisha and Deepali is 5:x. Manisha is 9 years younger than Parineeta. Parineeta's age after 9 years will be 33 years. The difference between Deepali's and Manisha's age is same as the present age of Parineeta. What will come in place of x?

(a) 23

(b) 39

(c) 15

(d) None of these

Ans: (d) Given Parineeta's age after 9 years = 33 years

$\therefore$ Parineeta's present age = $33 - 9 = 24$ years

$\therefore$ Manisha's present age = $24 - 9$

= 15 years

$\therefore$ Deepali's present age = $15 + 24$

= 39 years Hence, ratio between Manisha and Deepali = 15:39 = 5:13

$\therefore x = 13$

Example 11. The ratio between the present ages of Ram and Rakesh is 6:11. 4 years ago, the ratio of their ages was 1:2. What will be Rakesh's age after five

years?

(a) 45 years

(b) 29 years

(c) 49 years

(d) Cannot be determined

Ans: (c) Let, the age of Ram = x and, Rakesh = y, then, $\frac{x}{y}$

$=\frac{6}{11}$

$\therefore$ x $=\frac{6y}{11}$

According to the question,

$$\frac{x-4}{y-4}=\frac{1}{2}$$

2x - 8 = y - 4

$$2\times\frac{6y}{11}-8=y-4$$

$$\frac{12y}{11}-y=-4+8$$

$$\frac{y}{11}=4$$

Y=44 years

$\therefore$ Age of Rakesh after 5 years = 44 + 5 = 49 years

Example 12. The ratio between the present ages of Ram Rohan and Raj is 3:4:5. If the average of their present ages is 28 years then what will be the sum of the ages of Ram and Rohan together after 5 years?

(a) 45 years

(b) 55 years

(c) 52 years

(d) 59 years

Ans: (d) Let, the ages of Ram, Rohan and Raj be 3x, 4x and 5x respectively.

Then,

$$\frac{3x + 4x + 5x}{3} = 28$$

⇒ 4x = 28

⇒ $x = \frac{28}{4} = 7 years$

So, the present ages of Ram and Rohan together

= 3x + 4x = 7x = 7 × 7 = 49 years

Hence, the sum of the ages of Ram and Rohan together after 5 years

= 49 + 5 × 2 = 49 + 10 = 59 years

Example 13. In a family, mother's age is twice that of daughter's age. Father is 10 years older than mother. Brother is 20 years younger than his mother and 5 years older than his sister. What is the age of the father?

(a) 62 years

(b) 60 years

(c) 58 years

(d) 55 years

Ans: (b) Let, the age of the daughter be x. Then, age of brother = x + 5 years Therefore, age of mother = 2x years ∴ 2x - 20 = x + 5 ⇒ 2x - x = 5 + 20 ⇒ x = 25 years. Age of mother = 2x = 2 × 25 = 50 years Age of father = 50 + 10 = 60 years

Summary

The key concepts learned from this unit are: -

- We have learnt how to calculate age some years ago.
- We have learnt how to calculate present age.
- We have learnt how to calculate age some years hence.

Keywords

- Age.
- Present age.
- Ratio of ages.

Self Assessment

Q1.10, years ago, Mohan was thrice as old as Ram was but 10 years hence, he will be only twice as old as Ram. Find Mohan's present age.

A. 60 years
B. 80 years
C. 70 years

D. 76 years

Q2.The ages of Ram and Shyam differ by 16 years. 6 years ago, Mohan's age was thrice as that of Ram's, find their present ages.

A. 14 years, 30 years
B. 12 years, 28 years
C. 16 years, 34 years
D. 18 years, 38 years

Q3.In a family, mother's age is twice that of daughter's age. Father is 10 years older than mother. Brother is 20 years younger than his mother and 5 years older than his sister. What is the age of the father?

A. 62 years
B. 60 years
C. 58 years
D. 55 years

Q4.The ratio between the present ages of Ram Rohan and Raj is 3:4:5. If the average of their present ages is 28 years then what will be the sum of the ages of Ram and Rohan together after 5 years?

A. 45 years
B. 55 years
C. 52 years
D. 59 years

Q5.The age of a father is three times of that of his son. After 5 years, the double of father's age will be five times the age of son. The present age of father and son is:

A. 30 years, 10 years
B. 36 years, 12 years
C. 42 years, 14 years
D. 45 years, 15 years

Q6.The ratio between the present age of Manisha and Deepali is 5:x. Manisha is 9 years younger than Parineeta. Parineeta's age after 9 years will be 33 years. The difference between Deepali's and Manisha's age is same as the present age of Parineeta. What will come in place of x?

A. 23
B. 39
C. 15
D. None of these

Q7.After replacing an old member by a new member, it was found that the average age of fi ve members of a club is same as it was 3 years ago. The difference between the ages of the replaced and the new members is:

A. 2 years
B. 4 years
C. 8 years

D. 15 years

Q8.One year ago, a father was four times as old as his son. In 6 years time his age exceeds twice his son's age by 9 years. Ratio of their ages is:

A. 13:4

B. 12:5

C. 11:3

D. 9:2

Q9.The sum of the ages of a father and son is 45 years. 5 years ago, the product of their ages was four times the father's age at that time. The present age of the father is:

A. 39 years

B. 36 years

C. 25 years

D. None of these

Q10. Three times the present age of a father is equal to eight times the present age of his son. 8 years hence the father will be twice as old as his son at that time. What are their present ages?

A. 35, 15

B. 32, 12

C. 40, 15

D. 27, 8

Q11. The ratio of A's and B's ages is 4:5. If the difference between the present age of A and B 5 years hence is 3, then what is the sum of present ages of A and B?

A. 68 years

B. 72 years

C. 76 years

D. 64 years

Q12. The sum of the ages of A and B is 42 years. 3 years back, the age of A was 5 times the age of B. Find the difference between the present ages of A and B.

A. 24 years

B. 36 years

C. 72 years

D. 60 years

Q13. The age of father is 4 times the age of his son. If 5 years ago father's age was 7 times the age of his son at that time, then what is father's present age?

A. 24 years

B. 40 years

C. 72 years

D. 60 years

Q14. 10 years ago Anu's mother was 4 times older than her daughter. After 10 years, the mother will be twice older than her daughter. Find the present age of Anu is:

A. 24 years

B. 40 years

C. 20 years

D. 60 years

Q15. 6 years ago Mahesh was twice as old as Suresh. If the ratio of their present ages is 9:5 then, what is the difference between their present ages?

A. 24 years

B. 40 years

C. 20 years

D. 60 years

Answers for Self Assessment

1.	C	2.	A	3.	B	4.	D	5.	D
6.	D	7.	D	8.	C	9.	B	10.	B
11.	B	12.	A	13.	B	14.	C	15.	A

Review Questions

1. Two Ratio of the ages of Tania and Rakesh is 9:10. 10 years ago, the ratio of their ages was 4:5. What is the present age of Rakesh?
2. The ratio of the ages of a father and a son at present is 5:2. 4 years hence, the ratio of the ages of the son and his mother will be 1:2. What is the ratio of the present ages of the father and the mother?
3. The ratio of the ages of Anubha and her mother is 1:2. After 6 years the ratio of their ages will be 11:20. 9 years before, what was the ratio of their ages?
4. The ratio of the present ages of Swati and Trupti is 4:5. 6 years hence the ratio of their ages will be 6:7. What is the difference between their ages?
5. Radha's present age is three years less than twice her age 12 years ago. Also the ratio between Raj's persent age and Radha's present age in 4:9. What will be Raj's age after 5 years?
6. The ratio of the age of Tina and Rakesh is 9:10. 10 years ago the ratio of their ages was 4:5. What is the present age of Rakesh?
7. In a family, mother's age is twice that of daughter's age. Father is 10 years older than mother. Brother is 20 years younger than his mother and 5 years older than his sister. What is the age of the father?
8. The sum of the ages of two brothers, having a difference of 8 years between them, will double after 10 years. What is the ratio of the age of the younger brother to that of the elder brother?

9. After replacing an old member by a new member, it was found that the average age of fi ve members of a club is same as it was 3 years ago. The difference between the ages of the replaced and the new members is?
10. The age of a father is three times of that of his son. After 5 years, the double of father's age will be five times the age of son. The present age of father and son is?

Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle